Ecological Niche

It is the role, position a species has in its environment. A species niche include all of its interaction with biotic and abiotic factor of its environment.

Ecological Succession

It is a natural process by which different groups or communities colonise the same area over a period of time in a definite sequence. It can also be defined as series of stages one after the other, by which a group of organisms living in a community reaches its final stable state or climax.

POLLUTION

Introduction of contaminants or impurities into the natural environment is called pollution. Pollutant or contaminant may be natural or foreign substance. Pollution cause adverse changes in the nature and in long term affects the life on Earth.

Environmental Pollution

Environmental pollution is the effect of undesirable changes in our surroundings that have harmful effects on plants, animals and human beings.

Primary Pollutants

- A primary pollutant is an air pollutant emitted directly from a source. A secondary pollutant is not directly emitted as such, but forms when other pollutants (primary pollutants) react in the atmosphere.
- Sulphur dioxide, Nitrogen oxides, Carbon monoxide, CFCs, CO_2, Suspended Particulate Matter (SPM) etc are primary pollutants.

Secondary Pollutants

These are derived from primary pollutants. *The examples of secondary pollutants are as follow*

- Particulate matter formed from gaseous primary pollutants and compounds in photochemical smog, such as nitrogen dioxide.
- Ground level ozone (O_3) formed from NO and Volatile Organic Compounds (VOCs).
- Peroxyacetyl Nitrate (PAN) similarly formed from NO_2 and VOCs.

Kinds of Pollution

1. Air Pollution

Presence of contaminants released by human activities into the Earth's atmosphere having potential of causing harm to property or the precious lives of plants, animals or humans.

Causes and Sources

- Air pollution is caused by various type of pollutants. Petroleum refineries release poisonous gases like sulphur dioxide, nitrogen oxides etc.
- Dust is produced from cement factories and from stone crushers and hot mix plant releases suspended particulate matter more than five times the safety limits by industrial standards.
- The thermal power plant produce deadly pollutants such as fly ash, hydrocarbons etc. In urban areas, automobiles are the chief sources of pollution.
- The ever increasing traffic density has aggravated the existing problem of air pollution. Automobiles mainly produce pollutants like unburnt hydrocarbons, CO_2, NO_2, lead oxides etc.

Effects of Air Pollution

- **Effect on Human Health**
  - Irritation of eyes, nose, throat, damage to lungs when inhaled.
  - Acute and chronic asthma.
  - Brochitis and emphysema (as a result of synergy between SO_2 and suspended particulate matter).
  - Lung cancer.
- **Acid Deposition** The process by which acidic air pollutants, generally sulphur dioxide and nitrogen oxides are deposited on the Earth. Much of this deposition occurs when the pollutants condense in water and fall to the Earth as precipitation, known generally as acid rain. Acid deposition causes changes in the pH of water and soil, leading to a host of environmental problems.
- **Wet Deposition** It refers to acidic rain, fog and snow. If, the acid chemicals in the air are blown into areas where the weather is wet, the acids can fall to the ground in the form of rain, snow, fog or mist.
- **Dry Deposition** In areas where the weather is dry, the acid chemicals may become incorporated into dust or smoke and fall to the ground through dry deposition, sticking to the ground, buildings, homes, cars and trees.

2. Noise Pollution

Noise can be described as sound without agreeable musical quality or as an unwanted or undesired sound. Thus, noise can be taken as a group of loud, non-harmonious sounds or vibrations that are unpleasant and irritating to ear.

Sources of Noise Pollution

- The noise pollution has two sources, i.e. industrial and non-industrial. The industrial sources includes the noise from various industries and big machines working at a very high speed and high noise intensity.
- Non-industrial source of noise includes the noise created by road traffic, aircraft, railroads, construction, industry, noise in buildings and consumer products, loud speakers, sirens etc.

Effects of Noise Pollution

- It decreases the hearing efficiency in humans. There should be cool and calm atmosphere during the pregnancy. Unpleasant sounds make a lady of irritative nature. Sudden noise causes abortion in females.
- The noises are recognised as major contributing factors in accelerating the already existing tensions of modern living. These tensions result in certain disease like enhanced blood pressure or mental illness etc.
- Noise pollution damages the nervous system of animals. Animal loses the control of its mind.
- Loud noise is very dangerous to buildings, bridges and monuments. It creates waves which strike the walls and put the building in danger condition. It weakens the structure of buildings.

3. Radioactive Pollution

Radioactive pollution, like any other kind of pollution, is the release of radioactive elements into the environment. Radioactive element omits high energy waves which can cause mutation in the DNA of living organisms.

Causes of Radioactive Pollution

- There are many causes of radioactive pollution, which can significantly harm the environment.
- Production of nuclear weapons, decommissioning of nuclear weapons, mining of radioactive ore, coal ash, medical waste, nuclear power plants are important source of radioactive pollution.

Radioactive Waste Management

Following techniques are effective in radioactive waste management

- **Geological Disposal** This is effectively, the burying of radioactive material. Rooms are then excavated at the bottom of geological formation and radioactive material is stored here until it has decayed enough to not be dangerous any more.
- **Transmutation** Transmutation of radioactive waste is the process of consuming this radioactive waste and turning it into less harmful waste. This is currently not used very often due to high coasts.
- **Re-Use of Radioactive Waste** Some radioactive isotopes, such as strontium-90 and caesium-137 are able to be extracted for use in other industries such as food irradiation.

e-Waste

- It comprises of wastes generated from used electronic devices and household appliances, which are no longer suitable for their original intended use and are best suited only for recovery, recycling or disposal such as computers, mobiles or cell phones, personal stereos and large household appliances such as refrigerators, air-conditioners.
- **e-Waste Treatment** Environmentally sound e-waste treatment technology is identified at three levels, *which are as follow :*
  i. In this level treatment includes decontamination, dismantling and segregation.
  ii. This treatment includes shredding and special treatment processes like electromagnetic separation, eddy current separation and density separation using water.
  iii. This treatment includes recovery of metals and disposal of hazardous e-waste fraction, including plastics with flame retardants, CFCs, capacitors, mercury, lead and other items. All three levels of e-waste treatment are based on material flow.

4. Water Pollution

According to definition of WHO, "water pollution occurs when foreign materials either from natural or other sources are added to water supplies and may be harmful to life, because of their toxicity, reduction of normal oxygen level of water, aesthetically unsuitable effects and spread of epidemic diseases".

Sources of Water Pollution

Sources of water pollution are as follow :

- Industrial effluents.
- Industrial wastes derived from chemical industries, thermal power plants and nuclear power stations.
- Sewage and other waste.
- Agricultural discharges.

Types of Water Pollution

Water pollution can be classified as :

- **Surface Water Pollution** Rivers, lakes and ponds constitute surface water.
- **River Water Pollution** It poses a serious problem. Urban sewer drainage and industrial effluents are two major sources of river pollution. Soil erosion causes siltation of lakes.
- **Sea Water Pollution** It occurs near coastal waters due to dumping of pollutants. Pollution at deep sea is caused by leakage of oil due to ship wreckage. Oil spills have caused great harm to sea life in recent past.
- **Ground Water Pollution** Contamination of ground water takes place through pollutants like nitrates, phosphorus, potash, insecticides, pesticides etc. These pollutants reach into the ground water and make water polluted.
- **Effects of Water Pollution** The water we drink is a crucial component for healthy living. Clean water is essential for health. Use of contaminated water is a major reason for various water borne diseases like Cholera, Typhoid, Hepatits-B etc.

- The effects of water pollution are not only devastating to people, but also to animals, fish and birds. Polluted water is unsuitable for drinking, recreating, agriculture and industry. It diminishes the aesthetic quality of lakes and rivers. More seriously, contaminated water destroys aquatic life and reduces its reproductive ability. Eventually, it is a hazard to human health.

5. Thermal Pollution

- This pollution is the degradation of water quality by any process that changes ambient water temperature. A common cause of thermal pollution is the use of water as a coolant, storm water by power plants and industrial manufactures.
- When water used as a coolant is returned to the natural environment at a higher temperature, the change in temperature decreases oxygen supply and affects ecosystem composition. Thermal pollution is a crucial source of global warming.

6. Marine Pollution

- It refers to the emptying of chemicals or other particles into the ocean and its harmful effects. These chemicals enter into food chain when they are taken up by plankton and benthos.
- UNCLOS gives special attention to protection and preservation of marine environment. *It covers main sources of ocean pollution, which are as follow*
  - Land based and coastal activities
  - Continental shelf drilling
  - Potential sea bed mining
  - Ocean dumping
  - Vessel source pollution
  - Pollution from or through the atmosphere.

BIODIVERSITY

It refers to the variety and abundance of organisms living in a particular region. In other words, biodiversity is the variability among living and non-living organism and ecological complexes of which they are part, including diversity within and between species and ecosystems.

Levels of Biodiversity

Biodiversity can be observed at three levels, which are as follows :

i. **Genetic Diversity** It means the variation found in the genes within a species. Each individual of every species have different genetic composition. Within a species there may also be discrete populations with distinctive genes.

ii. **Species Diversity** It refers to the variety of species in a specific area. The diversity of species can be measured through its richness, abundance and types.

iii. **Ecosystem Diversity** The broad differences between ecosystem types and the diversity of habitats and ecological processes occurring within each ecosystem type constitute the ecosystem diversity.

Measurement of Biodiversity

Biodiversity is an important measure of ecosystem health. Biodiversity can be measured and monitored at several spatial scales.

- **Alpha Diversity** It is used to measure richness and evenness of individuals within a habitat unit.
- **Beta Diversity** Beta diversity is distinct from alpha diversity as it is expression of diversity between habitats.
- **Gamma Diversity** It is used to measure diversity of habitats within a landscape or region.
- **Point Diversity** It refers to diversity on the smallest scale such as the diversity of microhabitat or sample taken from within a homogenous habitat.

Biodiversity Hotspots

- A biodiversity hotspot is a bio-geographic region with a significant reservoir of biodiversity that is under threat from humans. This concept was put forward by Norman Myers.
- To qualify as a biodiversity hotspot, a region must meet two strict criterias. It must contain atleast 0.5% or 1500 species of vascular plants as endemic and it has to have lost atleast 70% of its primary vegetation.
- Around the world, 25 areas qualify under this definition. These sites support nearly 60% of the world's plant, bird, mammal, reptile and amphibian species, with a very high share of endemic species.

Threats to Biodiversity

- The unsustainable harvesting of natural resources, including plants, animals and marine species and the degradation or fragmentation of ecosystems through land conversion for agriculture, forest clearing etc. Invasive non-native or alien species being introduced to ecosystems to which they are not adapted.
- Pollution from chemical contaminants certainly poses a further threat to species and ecosystems. A changing global climate threatens species and ecosystems. The distribution of species is largely determined by climate, as the distribution of ecosystems and plant vegetation zones.

Conservation of Biodiversity

Conservation of biodiversity is important to

- prevent the loss of genetic diversity of a species.
- save a species from becoming extinct.
- protect ecosystems damage and degradation.

Conservation Strategies

Conservation effects can be grouped into following two categories, which are as follow

i. In situ (on site) conservation included to protection of plants and animals with their natural habitats or in protected areas. Protected areas are land or sea dedicated to protect and maintain biodiversity.

ii. Ex-situ (off site) conservation of plants and animals outside the natural habitat. These include botanical gardens, gene banks of seed, tissue culture and cryopreservation.

International Efforts to Protect the Biodiversity

Critical Ecosystem Partnership Fund (CEPF) is a global programme that provides funding and technical assistance to non-governmental organisations and other private sector partners to protect critical ecosystems. They focus on biodiversity hot-spots.

The Convention on Biological Diversity (CBD)

The Convention on Biological Diversity (CBD) is an international legally binding treaty. *The convention has three main goals, which are as follow :*

i. Conservation of biodiversity.

ii. Sustainable use of its components.

iii. Fair and equitable sharing of benefits arising from genetic resources.

Cartagena Protocol in Biosafety

- On 29th January, 2000, the Conference of the Parties (CoP) to the convention on biological diversity adopted a supplementary agreement to the convention known as the cartagena protocol on biosafety.
- The protocol seeks to protect biological diversity from the potential risks posed by living modified organisms resulting from modern biotechnology.

The Nagoya Protocol on Access and Benefit-Sharing

The Nagoya Protocol on Access to genetic resources and the fair and equitable sharing of benefits arising from their utilisation to the convention on biological diversity is an international agreement which aims at sharing the benefits arising from the utilisation of genetic resources in a fair and equitable way.

International Union for Conservation (IUCN)

- IUCN is an international organisation dedicated to finding pragmatic solutions to our most pressing environment and development challenges.
- The organisation publishes the IUCN red list compiling information from a network of conservation organisations to rate which species are most endangered.

The IUCN Red List Classification

Extinct *(EX)*	No individuals remaining.
Extinct in the Wild *(EW)*	Known only to survive in captivity.
Critically Endangered *(CR)*	Extremely high risk of extinction in the wild.
Endangered *(EN)*	High risk of extinction in the wild.
Vulnerable *(VU)*	High risk of endangerment in the wild.
Near Threatened *(NT)*	Likely to become endangered in near future.
Least Concern *(LC)*	Lowest risk, does not qualify for a 'more at risk category'.
Data Deficient *(DD)*	Data not enough to make an assessment of its risk of extinction.
Not Evaluated *(NE)*	Has not been evaluated against the criteria.

Man and Biosphere Programme (MAB)

- MAB of UNESCO was established in 1971 to promote interdisciplinary approaches to management, research and education in ecosystem conservation and sustainable use of natural resources.
- Earlier focus of MAB programme was protection of designated area, but in 1990's after Rio Summit focus shifted towards promoting interactions of mankind with nature in terms of sustainable living, income generation and reducing poverty.

MAB-Network

- Total membership has reached 669 biosphere reserves in 120 countries. It was created in 1971.
- Benefits gained from being part of network include access to a shared base of knowledge and incentives to integrate conservational practices.

Conservation of Biodiversity in India

Several important steps have been taken by the Indian Government including enactment of laws for in-situ and ex-situ conservation of endangered and valuable plants and animal, creation of biosphere reserves, national parks, sanctuaries, world heritage sites, zoological parks are some of the important steps in this direction which have been described briefly as below

Endangered Species of India

Birds	Great Indian Bustard, Forest Owlet, Vulture, Bengal Florican, Himalayan Quail, Siberian Crane
Mammals	Flying Squirrel, Red Panda, Pygmy Hog, Kondana Rat Snow Leopard, Asiatic Lion, One-Horned Rhinoceros
Reptiles	Gharial, Hawksbill Turtle, River Terrapin, Sispara Day Gecko
Amphibians	Flying Frog, Tiger Toad

Wildlife Conservation in India

Project	*Year*	*Project*	*Year*
Project Hangul	1970	Project Manipur Thamin	1977
Project Gir	1972	Project Rhino	1987
Project Tiger	1973	Project Elephant	1992
Project Olive Riddey Turtles	1975	Project Red Panda	1996
Crocodile Breeding Scheme	1975	Project Vulture	2006

Tiger Conservation

'The Project Tiger' was launched in India in 1972 as conservation programme for saving the Indian Tiger. Project Tiger Scheme has been under implementation since, 1973 as a Centrally Sponsored Scheme of Government of India.

Tiger Census 2014

The number of tigers in India has increased by 30% since, 2010 to 2226 in 2014 according to the Tiger Census, 2014 Karnataka has the highest number of tigers (406) followed by Uttarakhand (340), Madhya Pradesh (308), Tamil Nadu (229) and Maharashtra (190).

Important Tiger Reserves

Name of Tiger Reserve	*State*	*Name of Tiger Reserve*	*State*
Kaziranga	Assam	Periyar	Kerala
Manas	Assam	Dudhwa	Uttar Pradesh
Nameri	Assam	Simplipal	Odisha
Namdapha	Arunachal Pradesh	Ranthambore	Rajasthan
Nagarjunsagar	Andhra Pradesh-Telangana	Melghat	Maharashtra
Valmiki	Bihar	Sariska	Rajasthan
Indravati	Chhattisgarh	Pench	Madhya Pradesh
Bandipur	Karnataka	Bori, Satpura, Pachmari	Madhya Pradesh
Palamau	Jharkhand	Sathyamanglam	Tamil Nadu
Nagarhole	Karnataka	Kawal	Telangana
Corbett	Uttarakhand	Mukundra Hills	Rajasthan
Kanha	Madhya Pradesh		

Wetlands

- A wetland is a land area that is saturated with water, either permanently or seasonally, such that it takes on the characteristics of a distinct ecosystem.
- Wetlands consist primarily of hydric soil, which supports aquatic plants. Wetlands play a number of roles in the environment, principally water purification, flood control and shoreline stability.
- Wetlands are also considered the most biologically diverse of all ecosystems, serving as home to a wide range of plant and animal life.

Ramsar Convention

- The Ramsar convention is an international treaty signed in Ramsar, Iran for conservation and wise use of wetlands. The agreement was signed on 2nd February, 1971 and came into force from 21st December, 1975.
- It is one of the oldest specific conventions that deal not only with the conservation of the wetlands, but also its wise use. India joined Ramsar Convention in 1981 and this convention is in force in India since, 1982.

Ramsar Wetlands and their State

S. No.	*Wetland*	*State*
01.	Kolleru Lake	Andhra Pradesh
02.	Deepor Beel	Assam
03.	Nalsarovar	Gujarat
04.	Pong Dam Lake	Himachal Pradesh
05.	Renuka Wetland	Himachal Pradesh
06.	Chandertal Wetland	Himachal Pradesh
07.	Hokera Wetland	Jammu and Kashmir
08.	Surinsar-Mansar Lake	Jammu and Kashmir
09.	Tsomoriri	Jammu and Kashmir
10.	Wular Lake	Jammu and Kashmir
11.	Thrissur Kole	Kerala
12.	Vembanad-Kol Wetland	Kerala
13.	Sasthamkotta Lake	Kerala
14.	Ashtamudi Wetland	Kerala
15.	Bhoj Wetland	Madhya Pradesh
16.	Loktak Lake	Manipur
17.	Chilika	Odisha
18.	Bhitarkanika Mangrove	Odisha
19.	East Calcutta Wetlands	West Bengal
20.	Ropar	Punjab
21.	Kanjili	Punjab
22.	Harika Lake	Punjab
23.	Keoladeo National Park	Rajasthan
24.	Sambar Lake	Rajasthan
25.	Point Calimere Wildlife and Bird Sanctuary	Tamil Nadu
26.	Rudrasagar Lake	Tripura
27.	Upper Ganga River (*Brijghat to Narora Stretch*)	Uttar Pradesh